

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

BLOCK CHAIN TECHNOLOGY [CSE_4033]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the concept of Merkle Damgard Transform and its applicability in puzzle friendliness property of hash functions. Explain the process of commitment in blockchain networks and its significance in hiding mechanism. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Distinguish between decentralization and distribution in blockchain's record-keeping approach. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Apply hash functions to demonstrate how data integrity is maintained in blockchain transaction blocks. Select a simple example to illustrate how a puzzle-friendly hash function is used in Proof-of-Work mining. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Analyze the role of different consensus mechanisms in preventing double - spend attacks. Identify mechanisms by which an honest user can be protected from double - spend attacks. Suggest possible approaches (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Compare and contrast Proof of Work and Proof of Stake with examples. Evaluate their advantages and disadvantages. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Analyze different applications of Bitcoin scripts in real-world use cases (escrow, atomic swaps, time-locked transactions). (3)
- Course Outcome: "CO5" / Cognitive Level: "Analysis"
- C) Analyze how the components of Ethereum architecture - blockchain, EVM, and smart contracts - interact to enable decentralized application execution. (2)
- Course Outcome: "CO5" / Cognitive Level: "Analysis"
- 4) Apply the concept of Proof-of-Work to illustrate how Bitcoin mining leads to high energy consumption and ecological challenges. Propose suitable technological or policy-based solutions to mitigate these issues (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Develop and apply a conceptual model to demonstrate how identity management operates differently in public and permissioned blockchains, focusing on authentication, authorization, and trust mechanisms. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Utilize Bitcoin's address graph to demonstrate how clustering can identify related accounts. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Compare and contrast privacy in Bitcoin, Monero, and Zcash. (5)

- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Apply the idea of colored coins to represent smart property. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Explain the relationship between consensus on rules, history, and value. Describe the Silk Road case and its significance for Bitcoin regulation. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_4033:BLOCK CHAIN
TECHNOLOGY

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	10
3	3	CO 3	11
4	4	CO 4	7
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_4033:BLOCK CHAIN
TECHNOLOGY

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	6	23(46%)
4	Analysis	6	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		13	50 (100%)